

Data wrangling across columns

Agenda and reminders

- HW 2 due tonight on **GitHub classroom**
 - Commit early and often, let me know if you have any technical problems
 - Make sure to submit both the `.qmd` **and** `.md` files
- Today: data wrangling across columns

Warmup activity

Work on the activity (handout) with a neighbor, then we will discuss as a class

Warmup

```
1 diamonds |>
2   summarize(mean_carat = mean(carat),
3             sd_carat = sd(carat),
4             mean_depth = mean(depth),
5             sd_depth = sd(depth),
6             mean_price = mean(price),
7             sd_price = sd(price))
```

A tibble: 1 × 6

	mean_carat	sd_carat	mean_depth	sd_depth	mean_price	sd_price
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.798	0.474	61.7	1.43	3933.	3989.

Are there any downsides to this code?

Warmup

```
1 diamonds |>
2   summarize(mean_carat = mean(carat),
3             sd_carat = sd(carat),
4             mean_depth = mean(depth),
5             sd_depth = sd(depth),
6             mean_price = mean(price),
7             sd_price = sd(price))
```

- more variables to summarize means longer code, harder to read
- requires a lot of copying and pasting
- more prone to errors when typing names of functions, variables, etc.

across: Data wrangling across columns

Instead of copying the same function multiple times for different columns, we can apply functions *across* the columns of a table:

```
1 diamonds |>  
2   summarize(across(c(carat, depth, price),  
3                   mean))
```

```
# A tibble: 1 × 3  
  carat depth price  
  <dbl> <dbl> <dbl>  
1 0.798  61.7 3933.
```

across: Data wrangling across columns

Instead of copying the same function multiple times for different columns, we can apply functions *across* the columns of a table:

```
1 diamonds |>
2   summarize(across(c(carat, depth, price),
3                     mean))
```

```
# A tibble: 1 × 3
  carat depth price
  <dbl> <dbl> <dbl>
1 0.798  61.7 3933.
```

What if I want to calculate both the mean *and* the standard deviation of these columns?

across with multiple functions

```
1 diamonds |>
2   summarize(across(c(carat, depth, price),
3                     list(mean, sd)))
```

A tibble: 1 × 6

	carat_1	carat_2	depth_1	depth_2	price_1	price_2
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.798	0.474	61.7	1.43	3933.	3989.

What if I want to include the function name in the summary columns?

across with multiple functions

```
1 diamonds |>
2   summarize(across(c(carat, depth, price),
3                     list("mean" = mean, "sd" = sd)))
```

A tibble: 1 × 6

	carat_mean	carat_sd	depth_mean	depth_sd	price_mean	price_sd
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.798	0.474	61.7	1.43	3933.	3989.

What if I want to change the order of the column names (e.g. mean_carat vs. carat_mean)?

across with multiple functions

```
1 diamonds |>
2   summarize(across(c(carat, depth, price),
3                     list("mean" = mean, "sd" = sd),
4                     .names = "{.col}_{.fn}"))
```

A tibble: 1 × 6

	carat_mean	carat_sd	depth_mean	depth_sd	price_mean	price_sd
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.798	0.474	61.7	1.43	3933.	3989.

```
1 diamonds |>
2   summarize(across(c(carat, depth, price),
3                     list("mean" = mean, "sd" = sd),
4                     .names = "{.fn}_{.col}"))
```

A tibble: 1 × 6

	mean_carat	sd_carat	mean_depth	sd_depth	mean_price	sd_price
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.798	0.474	61.7	1.43	3933.	3989.

Summarizing more columns

```
1 diamonds |>
2   summarize(across(c(carat, depth, price),
3                     list("mean" = mean)))
```

A tibble: 1 × 3

	carat_mean	depth_mean	price_mean
	<dbl>	<dbl>	<dbl>
1	0.798	61.7	3933.

How would I modify this code to calculate the mean for *all* the numeric variables (carat, depth, table, price, x, y, z)?

Summarizing more columns

Option 1:

```
1 diamonds |>
2   summarize(across(c(carat, depth, table, price, x, y, z),
3                     list("mean" = mean)))
```

A tibble: 1 × 7

	carat_mean	depth_mean	table_mean	price_mean	x_mean	y_mean	z_mean
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.798	61.7	57.5	3933.	5.73	5.73	3.54

Are there any issues with this approach?

Efficiently summarizing more columns

```
1 diamonds |>  
2   summarize(across(where(is.numeric),  
3                   list("mean" = mean)))
```

A tibble: 1 × 7

	carat_mean	depth_mean	table_mean	price_mean	x_mean	y_mean	z_mean
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.798	61.7	57.5	3933.	5.73	5.73	3.54

Efficiently summarizing more columns

```
1 diamonds |>
2   summarize(across(where(is.numeric),
3                     list("mean" = mean)))
```

A tibble: 1 × 7

	carat_mean	depth_mean	table_mean	price_mean	x_mean	y_mean	z_mean
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.798	61.7	57.5	3933.	5.73	5.73	3.54

`where(is.numeric)` returns the columns which are numeric:

```
1 is.numeric(diamonds$carat)
```

```
[1] TRUE
```

```
1 is.numeric(diamonds$price)
```

```
[1] TRUE
```

```
1 is.numeric(diamonds$clarity)
```

```
[1] FALSE
```

Efficiently summarizing more columns

We can use `where` with other functions too. For example:

```
1 diamonds |>
2   summarize(across(where(is.factor),
3                     list("num_categories" = n_distinct)))
```

A tibble: 1 × 3

	cut_num_categories	color_num_categories	clarity_num_categories
	<int>	<int>	<int>
1	5	7	8

What do you think this code is doing?

Class activity

- Work with a neighbor on the class activity
- At the end of class, submit your work as an HTML file on Canvas (one per group, list all your names)